

{ } 01 interactive vs. static copy.jsonc > { } encoding > { } x > { } axis

```
1 {
2   "data": {
3     "name": "dataset"
4   },
```

```
5   "mark": {
6     "type": "bar"
7   },
```

```
8   "encoding": {
9     "y": {
10      "field": "Country",
11      "type": "nominal"
12    },
13    "x": {
14      "type": "quantitative",
15      "field": "Margin"
16    }
17  }
18 }
19 }
```

```
1 {
2   "data": {
3     "name": "dataset"
4   },
5+  "layer": [
6+    {
7+      "mark": {
8+        "type": "bar",
9+        "opacity": 0.3,
10+       "tooltip": true
11+      },
12+      "encoding": {
13+        "x": {
14+          "field": "Margin"
15+        }
16+      }
17+    },
18+    {
19+      "mark": {
20+        "type": "bar",
21+        "tooltip": true
22+      },
23+      "encoding": {
24+        "x": {
25+          "field": "Margin_highlight"
26+        },
27+        "opacity": {
28+          "condition": {
29+            "test": {
30+              "field": "__selected__",
31+              "equal": "off"
32+            },
33+            "value": 0
34+          },
35+          "value": 1
36+        }
37+      }
38+    }
39+  ],
40  "encoding": {
41    "y": {
42      "field": "Country",
43      "type": "nominal"
44    },
45    "x": {
46      "type": "quantitative",
47+     "axis": {
48+       "title": "Margin"
49+     }
50  }
51 }
52 }
```


{ } 02 How to start - Axis copy.jsonc > [] layer > { } 0 > { } mark > # opacity

```
1 {
2   "data": {
3     "name": "dataset"
4   },
5   "layer": [
6     {
7       "mark": {
8         "type": "bar",
9         "opacity": 0.3
10      },
11      "encoding": {
12-     "x": {
13         "field": "Margin"
14       }
15     },
16     {
17       "mark": {
18         "type": "bar"
19       },
20       "encoding": {
22-     "x": {
23         "field": "Margin_highlight"
24       },
25       "opacity": {
26         "condition": {
27           "test": {
28             "field": "__selected__",
29             "equal": "off"
30           },
31           "value": 0
32         },
33         "value": 1
34       }
35     }
36   ],
37   "encoding": {
39-   "y": {
40-     "field": "Country",
41-     "type": "nominal"
42-   },
43   "x": {
44-   "type": "quantitative",
45   "axis": {
46-   "title": "Margin"
47   }
48 }
49 }
50 }
```

```
1 {
2   "data": {
3     "name": "dataset"
4   },
5   "layer": [
6     {
7       "mark": {
8         "type": "bar",
9         "opacity": 0.3
10      },
11      "encoding": {
12+     "y": {
13         "field": "Margin"
14       }
15     },
16     {
17       "mark": {
18         "type": "bar"
19       },
20       "encoding": {
22+     "y": {
23         "field": "Margin_highlight"
24       },
25       "opacity": {
26         "condition": {
27           "test": {
28             "field": "__selected__",
29             "equal": "off"
30           },
31           "value": 0
32         },
33         "value": 1
34       }
35     }
36   ],
37   "encoding": {
38
39   "x": {
40+     "field": "Year Month",
41+     "type": "nominal",
42+     "timeUnit": "yearmonth",
43     "axis": {
44+     "labelAngle": 0,
45+     "format": "%m.%Y",
46+     "tickCount": 0,
47+     "labelFontSize": 8,
48+     "ticks": false,
49+     "title": null
50     }
51+   },
52+   "y": {
53+     "type": "quantitative",
54+     "axis": null
55   }
56 }
57 }
```


{ } 03 Matrix copy.jsonc > [] layer > { } 0 > { } mark

```
6   {
7     "mark": {
8       "type": "bar",
9       "opacity": 0.3
10    }
11  },
12  {
13    "mark": {
14      "type": "bar"
15    },
16    "encoding": {
17      "y": {
18        "field": "Margin_highlight"
19      },
20      "opacity": {
21        "condition": {
22          "test": {
```

✧ 8 hidden lines

```
31  },
32  {
33    "mark": {
34      "type": "text",
35      "size": 12,
36      "yOffset": -5
37    },
38    "encoding": {
39      "text": {
40        "field": "Margin",
41        "format": ",.3s"
```

```
42  }
43  }
44  },
45  ],
46  "encoding": {
```

```
47    "x": {
48      "field": "Year Month",
49      "type": "nominal",
50      "timeUnit": "yearmonth",
51      "axis": {
52        "labelAngle": 0,
53        "format": "%m.%Y",
54        "tickCount": 2,
55        "labelFontSize": 8,
56        "ticks": false,
57        "title": null
```

```
58    }
59  },
60  "y": {
61    "type": "quantitative",
62    "axis": null,
63    "field": "Margin"
```

```
64  }
```

```
65  }
```

```
66  }
```

```
67  }
```

```
68  }
```

```
6   {
7     "mark": {
8       "type": "rect",
9       "opacity": 0.3
10    }
11  },
12  {
13    "mark": {
14      "type": "rect"
15    },
16    "encoding": {
17      "y": {
18        "field": "Category",
19        "type": "nominal"
20      },
21      "opacity": {
22        "condition": {
23          "test": {
```

✧ 8 hidden lines

```
29  },
30  {
31    "mark": {
32      "type": "text"
33    },
34    "encoding": {
35      "text": {
36        "field": "Margin_highlight",
37        "format": ",.2s",
38        "opacity": {
39          "condition": {
40            "test": {
41              "field": "__selected__",
42              "equal": "off"
43            },
44            "value": 0.3
45          },
46          "value": 1
47        }
48      }
49    },
50  ],
51  "encoding": {
```

```
52    "y": {
53      "field": "Category",
54      "type": "nominal"
55    },
56    "x": {
57      "field": "Year Month",
58      "type": "nominal",
59      "timeUnit": "yearmonth",
60      "axis": {
61        "labelAngle": 0,
62        "format": "%m.%Y",
63        "tickCount": 2,
64        "labelFontSize": 10,
65        "ticks": false,
66        "title": null,
67        "orient": "top"
```

```
68    }
69  }
```

```
70  }
```

```
71  }
```

```
72  }
```

```
73  }
```

```
74  }
```

```
75  }
```

```
76  }
```

```
77  }
```


5 hidden lines | [] layer

```

6   {
7     "mark": {
8       "type": "bar",
9     "opacity": 0.3
10  },
11    "encoding": {
12      "y": {
13        "field": "Margin"
14      }
15    }
16  },
17  {
18    "mark": {
19      "type": "bar"
20    },
21    "encoding": {
22      "y": {
23
34    "value": 1
35    }
36  },
37  {
38    "mark": {
39      "type": "text",
40    "size": 12,
41    "yOffset": -5
42    },
43    "encoding": {
44      "text": { "field": "Margin",
45      "format": ",.3s" },
46      "y": {
47        "field": "Margin"
48      }

```

23 hidden lines

5 hidden lines | [] layer

```

6   {
7     "mark": {
8       "type": "bar",
9     "opacity": 0.1,
10    "color": {
11      "expr": "datum['Margin'] >= datum['Margin SPLY'] ? '#ec7629' : '#b7b7b7' //
12    }
13  },
14    "encoding": {
15      "y": {
16        "field": "Margin"
17      }
18    }
19  },
20  {
21    "mark": {
22      "type": "bar",
23    "color": {
24      "expr": "datum['Margin'] >= datum['Margin SPLY'] ? '#ec7629' : '#b7b7b7' //
25    }
26  },
27    "encoding": {
28      "y": {
39    "value": 1
40    }
41  },
42  {
43    "mark": {
44      "type": "text",
45    "size": {
46      "expr": "datum['Margin'] >= datum['Margin SPLY'] ? 12 : 0" // Conditional ex
47    },
48    "yOffset": -5
49  },
50  },
51    "encoding": {
52      "text": {
53        "field": "Margin",
54        "format": ",.3s"
55      },
56      "y": {
57        "field": "Margin"
58      }

```

23 hidden lines

{} 05 Tooltip & Titles copy.jsonc > [] layer > {} 0

✧ 3 hidden lines | {} data

```
4   },
5   "layer": [
6     {
```

```
7       "mark": {
8         "type": "bar",
9         "opacity": 0.1,
```

✧ 27 hidden lines | {} 1 > {} encoding

```
37       "value": 0
38     },
39     "value": 1
40-  }
```

```
41  }
42  },
43  {
```

✧ 38 hidden lines

✧ 3 hidden lines | {} data

```
4   },
5   "layer": [
6     {
7+    "title": {
8+      "text": "Margin",
9+      "fontSize": 16,
10+     "anchor": "start",
11+     "color": "#454545",
12+     "subtitle": " vs PY ",
13+     "subtitleColor": "#454545"
14+   },
```

```
15     "mark": {
16       "type": "bar",
17       "opacity": 0.1,
```

✧ 27 hidden lines | {} 1 > {} encoding > {} opacity

```
45     "value": 0
46   },
47   "value": 1
```

```
48+ },
49+ "tooltip": [
50+   {
51+     "field": "Margin",
52+     "type": "quantitative",
53+     "format": ",.0f"
54+   },
55+   {
56+     "field": "Margin SPLY",
57+     "type": "quantitative",
58+     "format": ",.0f"
59+   }
60+ ]
```

```
61  }
62  },
63  {
```

✧ 38 hidden lines


```
4   },
5-  "layer": [
6-  {  "title": {
7-    "text": "Margin",
8    "fontSize": 16,
9-    "anchor": "start",
10   "color": "#454545",
11-   "subtitle": " vs PY ",
12   "subtitleColor": "#454545"
13  },
```

```
14    "mark": {
15      "type": "bar",
16      "opacity": 0.1,
```

29 hidden lines | {} 1 > {} encoding > {} opacity

```
4   },
5+  "title": {
6+    "text": "Monthly KPIs",
7    "fontSize": 16,
8+    "anchor": "middle",
9    "color": "#454545",
10+   "subtitle": " ",
11   "subtitleColor": "#454545"
12  },
13+  "vconcat": [
14+    {
15+      "title": {
16+        "text": "Margin",
17+        "anchor": "start",
18+        "color": "#454545",
19+        "subtitle": "Orange when CY > PY",
20+        "subtitleColor": "#454545",
21+        "offset": 15
22+      },
23+      "width": 400,
24+      "height": 180,
25+      "layer": [
26+        {
27          "mark": {
28            "type": "bar",
29            "opacity": 0.1,
```

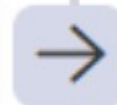
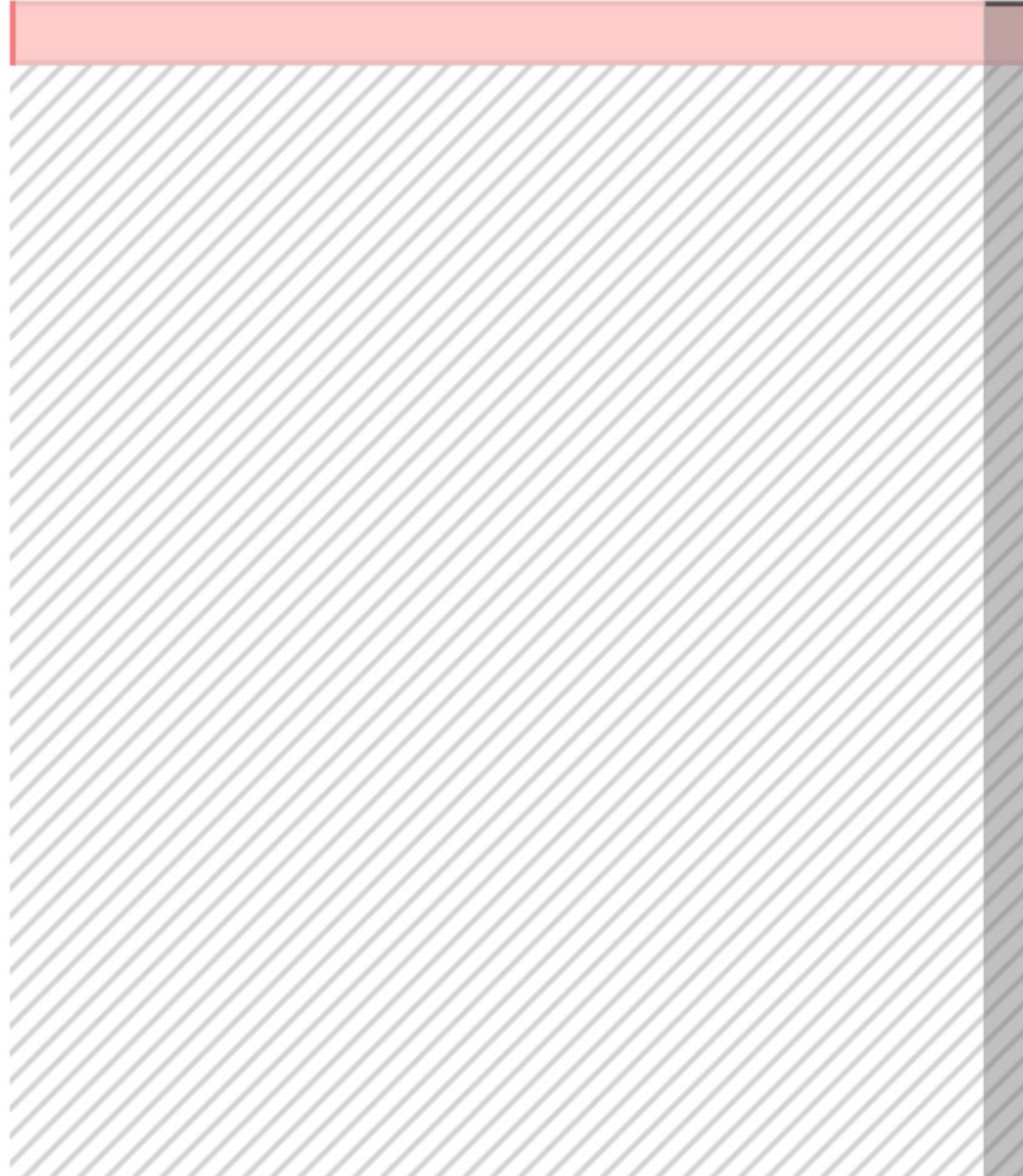
29 hidden lines | {} 1 > {} encoding > {} opacity

```
94   },
95   "y": {
96     "type": "quantitative",
97     "axis": null
98   }
99 }
```

```
107   },
108   "y": {
109     "type": "quantitative",
110     "axis": null
111   }
112 }
113+ },
114+ {
115+   "title": {
116+     "text": "Sales Amount",
117+     "anchor": "start",
118+     "color": "#454545",
119+     "subtitle": " ",
120+     "subtitleColor": "#454545",
121+     "offset": 15
122+   },
123+   "width": 400,
```


{ } 07 1 Config copy.jsonc > ...

1-



```
1+ {
2+   "view": {
3+     "stroke": false
4+   },
5+   "font": "Segoe UI",
6+   "text": {
7+     "fontSize": 11,
8+     "color": "#454545"
9+   },
10+  "axis": {
11+    "domain": false,
12+    "labelFontSize": 16,
13+    "labelFontWeight": 400
14+  },
15+  "axisX": {
16+    "labelPadding": 5
17+  }
18+ }
```


08 Repeat copy.jsonc > ...

```
1 {
2   "data": {
3     "name": "dataset"
4   },
```

```
5   "layer": [
6     {
7       "mark": {
```

5 hidden lines

```
13     },
14     "encoding": {
15       "y": {
16         "field": "Margin"
17       }
18     }
19   },
```

6 hidden lines | {} 1 > {} mark

```
26     },
27     "encoding": {
28       "y": {
29         "field": "Margin_highlight"
30       }
31     },
32     "opacity": {
33       "condition": {
```

6 hidden lines

```
39       "value": 1
40     },
41     "tooltip": [
42       {"field": "Margin", "type": "quantitative", "format": ",.0f"},
43       {"field": "Margin_SPLY", "type": "quantitative", "format": ",.0f"}
44     ]
45   }
46 },
```

7 hidden lines | {} 2 > {} mark

```
54     },
55     "encoding": {
56       "text": {
57         "field": "Margin",
58         "format": ",.3s"
59       },
60       "y": {
61         "field": "Margin"
62       }
63     }
64   }
```

14 hidden lines | {} encoding > {} x

```
79     },
80     "y": {
81       "type": "quantitative",
82       "axis": null
83     }
84   }
```

```
1 {
2   "data": {
3     "name": "dataset"
4   },
5+   "repeat": [
6+     "Margin",
7+     "Total Cost",
8+     "Sales Amount"
9+   ], // Added "repeat" for creating multi
10+   "columns": 1, // Specifies the layout of
11+   "spec": {
12+     "width": 450, // Sets the width of ea
13+     "height": 150, // Sets the height of
14   "layer": [
15     {
16       "mark": {
```

5 hidden lines

```
22     },
23     "encoding": {
24       "y": {
25+       "field": {
26+         "repeat": "repeat"
27+       }
28     }
29   },
30 },
```

6 hidden lines | {} 1 > {} mark

```
37     },
38     "encoding": {
39       "y": {
40+       "field": {
41+         "repeat": "repeat"
42+       }
43     },
44     "opacity": {
45       "condition": {
```

6 hidden lines

```
52       "value": 1
53     },
54     "tooltip": [
55+     {
56+       "field": {
57+         "repeat": "repeat"
58+       },
59+       "type": "quantitative",
60+       "format": ",.0f"
61+     }
62     ]
63   }
64 },
```

7 hidden lines | {} 2 > {} mark

```
72     },
73     "encoding": {
74       "text": {
75+       "field": {
76+         "repeat": "repeat"
77+       },
78       "format": ",.3s"
79     },
80     "y": {
81+     "field": {
82+       "repeat": "repeat"
83+     }
84   }
85 }
86 },
```

14 hidden lines | {} encoding > {} x

```
101     },
102     "y": {
103       "type": "quantitative",
104+     "axis": {
105+       "labelFontSize": 0,
106+       "ticks": false,
107+       "domain": false,
108+       "grid": false,
109+       "titleFontSize": 14
110+     }
111+   }
112 }
113 },
```


{} 09 Marking it up - Params copy.jsonc > {} layer

16 hidden lines | {} layer > {} 0 > {} encoding > {} y

17

}

18

}

19

},

20

{

21

"mark": {

22

"type": "bar",

23

"color": {

24

"expr": "datum['Margin']>= datum["

25

}"

26

},

27

"encoding": {

28

"y": {

29

}

30

},

31

"stroke": "#454545"

32

},

33

"tooltip": [

34

{"field": "Margin", "type": "quantita

35

]

36

}

8 hidden lines | {} opacity > {} condition

37

"value": 0

38

},

39

"value": 1

40

},

41

"tooltip": [

42

{"field": "Margin", "type": "quantita

43

]

44

}

43 hidden lines

16 hidden lines | {} layer > {} 0 > {} encoding > {} y

17

}

18

}

19

},

20

{

21

"params": [// Added "params" section for interaction configurations

22

{

23

"name": "jeff",

24

"select": {

25

"type": "point",

26

"on": "pointerover",

27

"clear": "pointerout"

28

},

29

{

30

"name": "sue",

31

"select": "interval"

32

}

33

],

34

"mark": {

35

"type": "bar",

36

"color": {

37

"expr": "datum['Margin']>= datum['Margin SPLY'] ? '#ec7629' : '#b7b7b7'" //

38

},

39

"stroke": "#454545"

40

},

41

"encoding": {

42

"y": {

43

}

44

},

45

"strokeWidth": { // Added "strokeWidth" encoding

46

"condition": [

47

{

48

"param": "jeff",

49

"empty": false,

50

"value": 3

51

},

52

{

53

"param": "sue",

54

"empty": false,

55

"value": 3

56

}

57

],

58

"value": 0

59

},

60

"tooltip": [

61

{"field": "Margin", "type": "quantitative", "format": ",.0f"},

62

]

63

}

8 hidden lines | {} opacity > {} condition

51

"value": 0

52

},

53

"value": 1

54

},

55

"strokeWidth": { // Added "strokeWidth" encoding

56

"condition": [

57

{

58

"param": "jeff",

59

"empty": false,

60

"value": 3

61

},

62

{

63

"param": "sue",

64

"empty": false,

65

"value": 3

66

}

67

],

68

"value": 0

69

},

70

"tooltip": [

71

{"field": "Margin", "type": "quantitative", "format": ",.0f"},

72

]

73

}

43 hidden lines

→


```
54     },
55     {
```

```
56     "mark": {
57       "type": "text",
58       "size": {
59         "expr": "datum['Margin'] >= datum['Margin
        SPLY'] ? 12 : 0" // Conditional expression
        for size
60       },
61       "yOffset": -5
62     },
63     "encoding": {
64       "text": {
65         "field": "Margin",
66         "format": ",.3s"
67       },
68       "y": {
69         "field": "Margin"
```

24 hidden lines

```
54     },
55     {
```

```
56+     "transform": [
57+       {
58+         "window": [
59+           {
60+             "op": "rank",
61+             "as": "rank"
62+           }
63+         ],
64+         "sort": [
65+           {
66+             "field": "Margin",
67+             "order": "descending"
68+           }
69+         ]
70+       },
71+       {
72+         "filter": "datum.rank <=1"
73+       },
74+       {
75+         "calculate": "datum['Month Name'] + ' -
        +         had the highest - margin with ' + format
        +         (datum['Margin'], ',.4s')",
76+         "as": "NewText"
77+       }
78+     ],
79     "mark": {
80       "type": "text",
81+       "yOffset": -35,
82+       "lineBreak": "-",
83+       "fontSize": 11,
84+       "fontWeight": 600
85     },
86     "encoding": {
87       "text": {
88+         "field": "NewText"
89       },
90       "y": {
91         "field": "Margin"
```

24 hidden lines